

Mohamed Abdelrahman

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EDUCATION

University of Houston

Bachelor of Science in Computer Science - Major GPA: 3.50

Aug 2023 – May 2026

Houston, TX

PROJECTS

TallyCart | Django, EasyOCR, Prompt Engineering

- Designing Django backend to structure products data for optimized shopping recommendations
- Building relational data models and logic to parse and organize large volumes of receipt-derived data
- Integrating EasyOCR with LLM-powered postprocessing to convert raw receipt text into structured product records

C++ Ordered Map Library | C++17, CMake, LCOV, Valgrind, Catch2, GitHub Actions, TDD

- Developed high-performance ordered map data structure combining doubly linked list with hash table for O(1) lookups while preserving insertion order
- Implemented comprehensive testable codebase achieving 97.3% code coverage across 2,246 test assertions with Catch2 for reliable software development
- Designed maintainable STL-compatible interface with iterator support and exception safety through modern C++17 features
- Established CI/CD pipeline with automated testing through GitHub Actions and CMake, following software development lifecycle best practices
- Verified zero memory leaks through Valgrind analysis and created comprehensive documentation

Nota | Flutter, Dart, Bloc, SQFlite, Quill, Material 3

- Developed rich-text note-taking app for Android with offline storage through SQL queries in SQFlite
- Integrated Flutter Quill editor to support formatting such as bold, italics, headers, links, and lists
- Implemented labeling and filtering features for efficient note organization and retrieval
- Managed UI and business logic with Bloc state management for clean separation of concerns
- Designed responsive UI with Material 3 components and support for light/dark theme switching

EXPERIENCE

Director of Competitions

Jan 2025 – May 2025

Code[Coogs] - University Of Houston

Houston, TX

- Organized and hosted 9 competitive programming events teaching data structures, algorithms, and complexity analysis to 30+ students
- Arranged algorithmic contests on Kattis, curating problem sets aligned with LeetCode-style interview topics
- Hosted interactive workshops with live coding and peer discussions to improve student engagement
- Collaborated with student leaders and faculty to expand outreach and iterate on event formats

Software Engineer (Freelance)

Aug 2023 – Jan 2025

Outlier.ai

Remote

- Developed solutions for fine-tuning code generation models with Python and C++, improving quality by 40% based on LLM evaluation rubrics
- Engineered prompts to generate model deviations, enhancing the quality of model training data.
- Reviewed submissions for other contributors and provided feedback, improving task quality by 30%

Software Engineering Intern

Aug 2022 – Jan 2023

bld.ai

Remote

- Collaborated with five interns to develop a full-stack application with React, Django, and PostgreSQL
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month, following a two-week sprint cycle
- Adhered to Agile Practices with sprint planning, daily stand-ups, and retrospectives optimizing workflow by 50%

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart

Frameworks: Django, React, NextJS, Flutter

Tools: Git, Docker, Postman, Coverage, GitHub Actions, VS Code, Ubuntu Linux, Bash